GridRescue AI — Disclosure Statement

Team Secure AI Grid · Global Energy Hackathon 2026 · Clean Energy Solutions track · Brief 12 “Relocatable solar for a hospital on the move”

*This statement discloses what existed before the hackathon sprint, what was built during it, the AI tools used, the open-source software and open data relied on, and the limits of the results. Items marked **[TO CONFIRM]** are to be confirmed by the team before submission. No dates appear in this document by design; versions are labelled only.*

(i) Team and submission

- **Team:** Secure AI Grid. **Authors:** Pavan Manjunath (ORCID 0000-0002-2694-8807) — product, AI and engineering; Thomas Prüfer — energy systems and the business case.
- **Submission:** Brief 12, “Relocatable solar for a hospital on the move”, Clean Energy Solutions track, Global Energy Hackathon 2026. Worked example: National Orthopaedic Hospital, Enugu (Abakpa Junction, Abakaliki Road, Enugu).
- **Repository:** https://github.com/pavangemmanjunath-cell/GridRescue_AI — source code under the Apache License 2.0; NOTICE lists third-party material; CITATION.cff and AUTHORS.md name the authors.
- **Submission package (FINAL_DOC):** POC summary and one-pager (DOCX + PDF), 5-slide deck (PPTX + PDF), demo video and pitch video (MP4, each under 5 minutes), Business Model Canvas (PDF + PNG), financial model (XLSX + PDF) and this statement (DOCX + PDF).

(ii) What existed before the hackathon sprint

Pre-existing before the sprint: the GridRescue AI web-application frame and the 3D scene kit. **[TO CONFIRM by the team before submission]** No customer, patient or personal data was used at any point; the hospital's facts come from Problem Brief 12 and the public sources in (vi).

(iii) What was built during the hackathon

The statement shown in the demo applies: “This workflow was built during the Global Energy Hackathon 2026.” The four modules:

- **Grid mapper.** Sun, wind, water and grid lenses from open data; substations and lines within 50 and 100 km from OpenStreetMap; the published supply hours and tariff beside the pin; and the “hospital is moving” new-site lens, which ranks areas within 12 km on grid access and distance.
- **Relocation plan.** The brief's five starter questions answered with the site's numbers: two sites (today and the new building), a theatres-first load list, a movable energy system, the energy type (every energy-system family simulated; a grid-first fallback with UPS on the theatres when the full energy system is not yet affordable), the no-sun case, the move plan, money (5- and 10-year cost model, ownership options, funders), impact and data sources.
- **3D hour-by-hour outage simulation.** The hospital's site in 3D with hourly dispatch (solar and wind first, then the backup energy system, then the generator; the critical load is never shed); four stress tests (12 h evening outage, 24 h national grid collapse, worst solar month, 18 h outage); the outage-survival curve (the outage started at each hour of the day, for 4 to 24 h); and the **ML classification of the energy system** — a Random Forest of 30 CART decision trees written in pure Python (no ML library), trained on 10,000 synthetic sites labelled by the engine (8,000 for training, 2,000 held out), with 80.4 % agreement with the engine on the held-out sites (top-2 94.5 %; a single tree 76.9 %; most-common-family baseline 46.5 %). The hourly simulation verifies every pick; in the app the pick is shown with its share of the vote (76 % in the worked example).
- **ESG report.** Scope 1 and Scope 2 (GHG Protocol), E·S·G report lines and the carbon value at the buyer's price (marked potential, not issued), written only after the simulation has verified the design; generated for any site (checked for Enugu and Kano).

Also built: offline mode (GR_OFFLINE=1, bundled snapshots); one-click start (START.bat / start.sh); tests and CI — 63 backend tests, 23 frontend unit tests and 10 Playwright GUI specs, with CI on every push (backend tests on Python 3.11–3.13, lint, type-check, unit tests, production build, CITATION.cff validation, a start-script smoke test); the research-repository files (README, CITATION.cff, AUTHORS, DATA_CARD, MODEL_CARD, CONTRIBUTING, CODE_OF_CONDUCT, SECURITY, SUPPORT, ROADMAP, CHANGELOG, docs/, benchmarks/, examples/quickstart.py); and the two videos and the documents listed in (i). Other pages of the same code base (Advisor, Facility, Compare, Costs, Report, and Ask GridRescue — extractive retrieval with citations over a local SQLite full-text index, no language model) fall under the **[TO CONFIRM]** in (ii) as to what predates the sprint.

(iv) AI tools used

- An AI coding assistant (Claude, Anthropic) was used for code, documentation, the video pipeline and the narration scripts. **[TO CONFIRM wording with the team]**
- The voices in the demo and pitch videos are synthetic (Windows text-to-speech). **[TO CONFIRM]**
- No language model runs inside the product: Ask GridRescue answers by retrieval only; the energy-system classifier is the team's own Random Forest; every engineering figure comes from the deterministic engine. The authors reviewed the output of these tools and take responsibility for all content.

(v) Open-source software and licences

The repository's own code and compiled data files are under the Apache License 2.0 (LICENSE, NOTICE). Dependencies are installed by pip and npm, not vendored; licences as recorded in docs/reproducibility.md:

- **Backend:** FastAPI (MIT), uvicorn (BSD-3-Clause), httpx (BSD-3-Clause), pydantic and pydantic-settings (MIT), diskcache (Apache-2.0), python-dotenv (BSD-3-Clause); pytest, pytest-asyncio and respx for tests (MIT / Apache-2.0 / BSD-3-Clause). The classifier and the full-text index use only the Python standard library (no scikit-learn; SQLite FTS5).

- **Frontend:** React and react-dom (MIT), react-router-dom (MIT), Vite and vitest (MIT), TypeScript (Apache-2.0), three.js with @react-three/fiber and @react-three/drei (MIT), Leaflet (BSD-2-Clause), **react-leaflet (Hippocratic License 2.1** — noted in NOTICE and docs/limitations.md; it adds conditions on use, and anyone redistributing a built frontend should review it), recharts (MIT), zustand (MIT), @mediapipe/tasks-vision (Apache-2.0), @playwright/test (Apache-2.0). lightningcss (MPL-2.0) is a build tool and is not shipped. Map tiles come from tile.openstreetmap.org under the OSM Foundation tile usage policy; the solar-panel texture is a cropped NASA photograph in the public domain.

(vi) Datasets and open data — what is published, reported and modelled

- **Published (source named):** diesel ₦3,277 a litre, up 86 % in a year — NBS Automotive Gas Oil price watch, national average, not an Enugu pump price; grid tariff ₦160/kWh — EERC Band A, Enugu State market; sun 5.48 peak-sun hours a day, confirmed by NASA POWER, PVGIS (EU JRC), NLR PVWatts and Global Solar Atlas; Open-Meteo (ERA5, GloFAS, elevation, geocoding; CC BY 4.0); OpenStreetMap substations, lines, plants, health facilities and rivers (ODbL, © OpenStreetMap contributors; a Geofabrik Nigeria extract is bundled as backend/app/data/osm_ng.json; Overpass and Nominatim are read live); World Bank Indicators; ENERGYDATA.INFO; tariff tables for Nigeria, Kenya, South Africa and Ghana from NERC, EERC, EPRA, Eskom and PURC; market statistics from WHO/World Bank/IRENA/SEforALL ("Energizing health"), the World Bank Enterprise Surveys and the IEA Africa Energy Outlook.
- **Reported (Problem Brief 12 / the hospital's head engineer):** about 6 h of grid a day; about 1,550 kW connected; 350 beds; sensitive equipment damaged by unstable supply; the theatre waits 10–30 s for the generator at each grid drop; the move to a new building in about five years is the brief's scenario.
- **Modelled (engine, ±30 %):** grid class D "Poor" and 9 substations within 50 km (engine read of OpenStreetMap and the published supply hours); loads from bed-based benchmarks (critical 303 kW, protected 641 kW); the design (1,670 kWp ground-mounted solar, a 5,020 kWh containerised energy system, 500 kVA online UPS, the existing 1,000 kVA generator last); diesel litres, costs, payback and CO_2; the new-building pin is illustrative.
- **Training data:** 10,000 synthetic sites generated by the training script and labelled by the engine — simulated, not real, regenerated on demand. No personal, patient or customer data is used or stored.

(vii) Limitations and honesty notes

- Results are pre-feasibility estimates (about ±30 % on cost; the band is assumed, not derived from validation) — not an engineering design, a quotation or a guarantee.
- "100 % of the critical load" holds in the modelled outages (the four stress tests and the 24-hour cases), not as a universal guarantee. One typical 24-hour design day represents the year; outages longer than a day are approximated by the 24 h national-collapse case.
- "0 ms switchover" is the online UPS's modelled behaviour, not a measured value.
- Loads are bed-based benchmarks until the hospital's load audit; the land area (20,000 m²), the generator size (1,000 kVA) and the hospital's diesel bill are assumed and flagged "to confirm". The break-even bill of ₦117.9 M a month is therefore a condition, not a claim: the hospital saves from month one only if its real bill is above it.
- The classifier is validated against the engine only (80.4 % held-out agreement on synthetic sites), not against installed systems; rare families (wind-only, flow, hydrogen) are effectively never picked; the simulation checks every pick.
- Offline mode gives slightly different figures (845,654 L of diesel and 2,344 t of CO_2 avoided a year, against 866,678 L and 2,246 t with live open data) because the offline hourly weather is modelled.
- No production hardening: development servers only; no authentication, roles, rate limiting or audit log; public OpenStreetMap tiles are not meant for heavy use.
- Money: the software prices (US$ 2,000 assessment, US$ 3,000 licence per site-year) are proposed and need market validation; the US$ 150,000 ask, the site counts and the months are planning targets, not achievements or commitments; the energy-as-a-service fee (₦69.1 M a month) is the hospital's cost, not GridRescue AI revenue; investor return is not computed.
- The simulation itself is not claimed as new. REopt (NREL, free) sizes PV, a backup energy system and a generator so a site survives a specified outage — US-based with guidance for international sites, built for energy experts; HOMER Pro (UL Solutions, commercial) is annual microgrid simulation and least-cost design, priced for consultants; the HOMER Powering Health Tool (World Bank/ESMAP, free) gives least-cost solar + backup energy system + diesel combinations for clinics and hospitals — a sizing and cost tool, not an outage-by-outage simulation, with no relocation, no grid mapping and no ESG report; Odyssey Energy Solutions covers project planning, financing, procurement and monitoring for developers; MapYourGrid / the OpenStreetMap community maps grid infrastructure (our data partner, not a competitor). What is claimed is the integrated, hospital-specific, relocatable, financeable workflow — what none of them does in one flow:
  - (1) From a map pin to an investment-ready hospital plan in minutes, on open data, with no API keys and no expert.
  - (2) The grid mapper with the "hospital is moving" new-site lens: substations and lines within 50 and 100 km, published supply hours, areas ranked for the new building — then a relocation plan with a move cost and a reuse percentage.
  - (3) Every energy-system family simulated hour by hour on the site's own sun, grid hours and load, four stress tests, with a vendor-neutral ML classification of the energy system that the simulation verifies.
  - (4) Local market assumptions built in: the NBS diesel price, the state tariff, Nigerian funders and programmes.
  - (5) The ESG report (Scope 1 and Scope 2, E·S·G lines) and the before → after KPIs written before anything is bought.

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